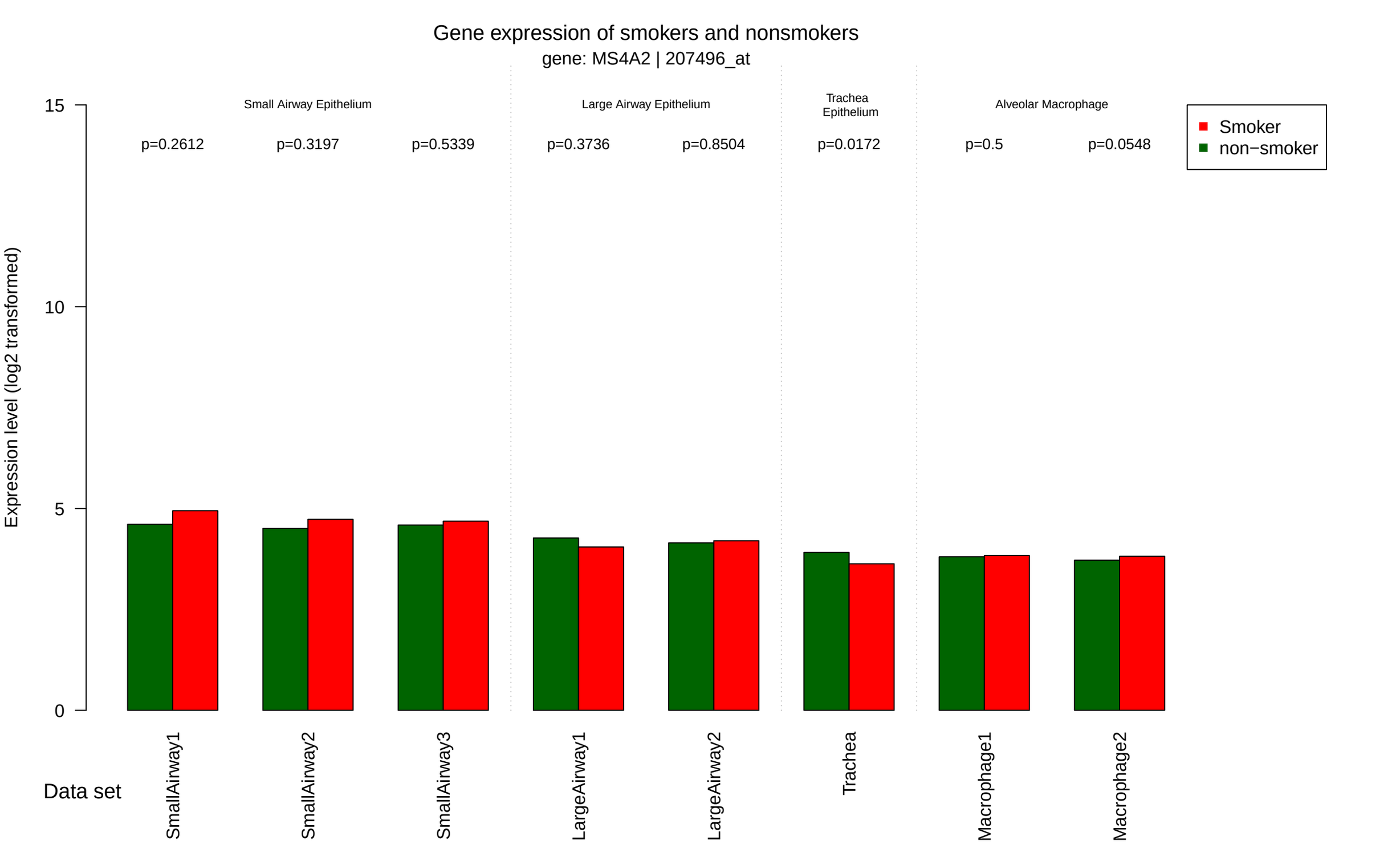
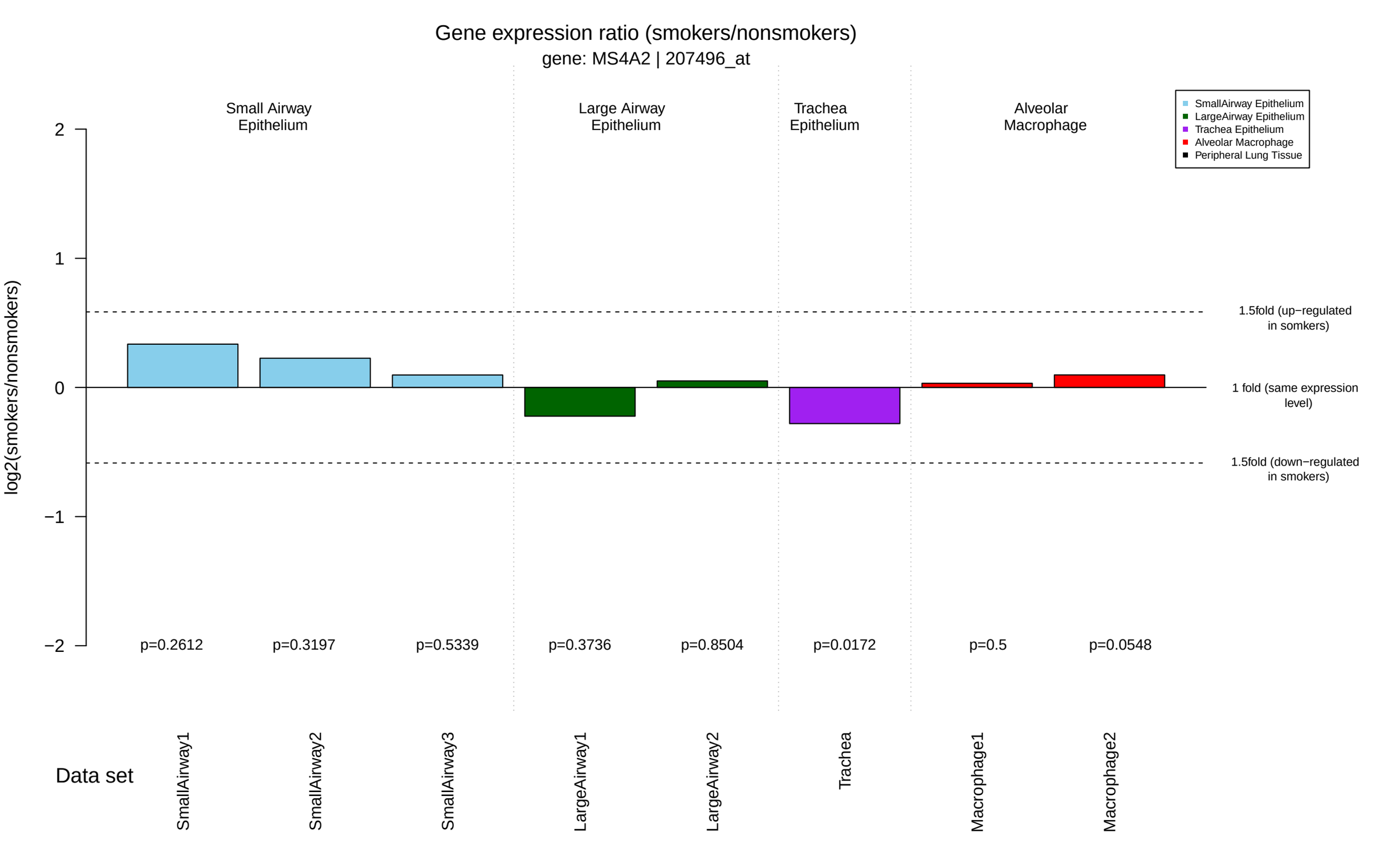
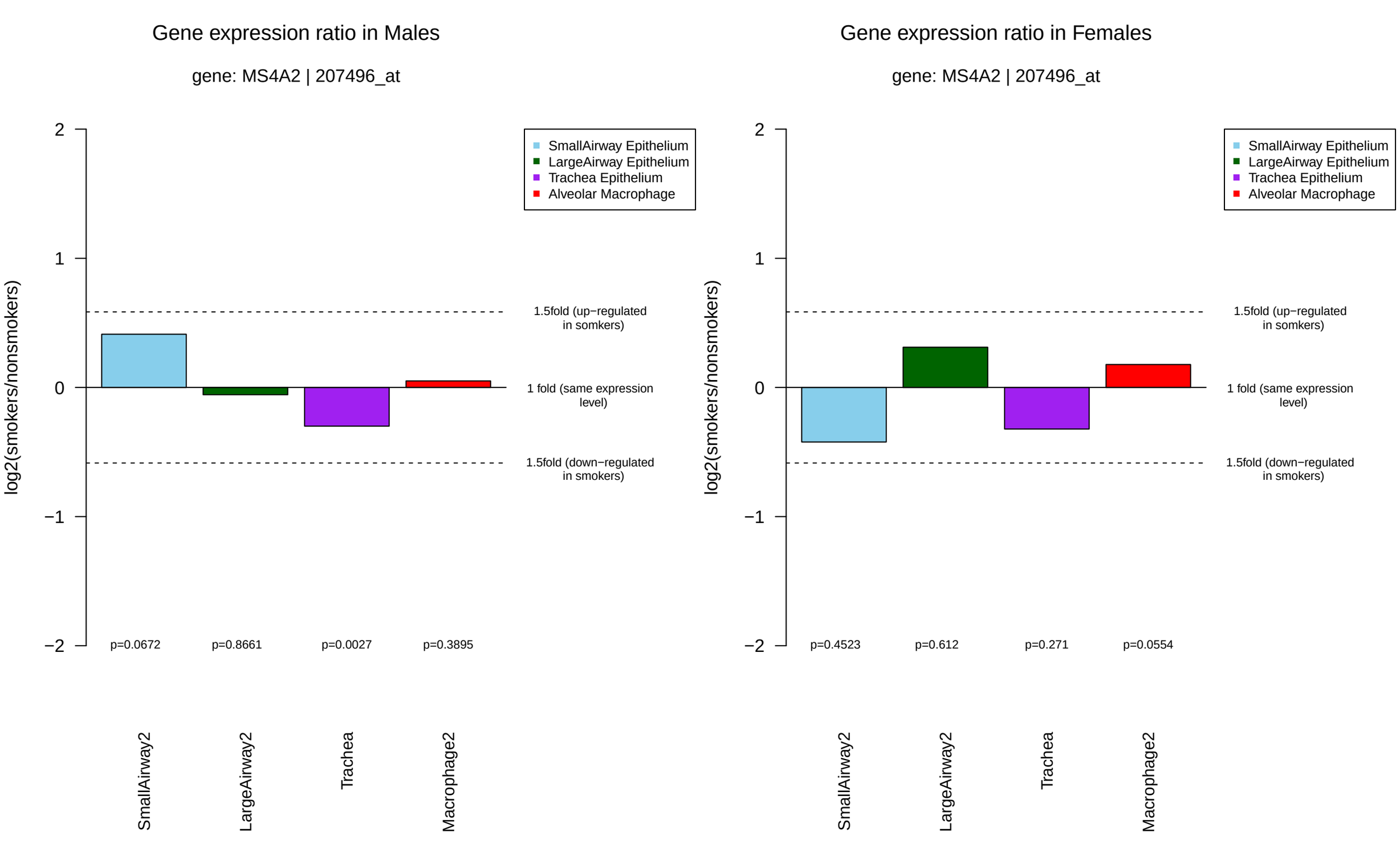
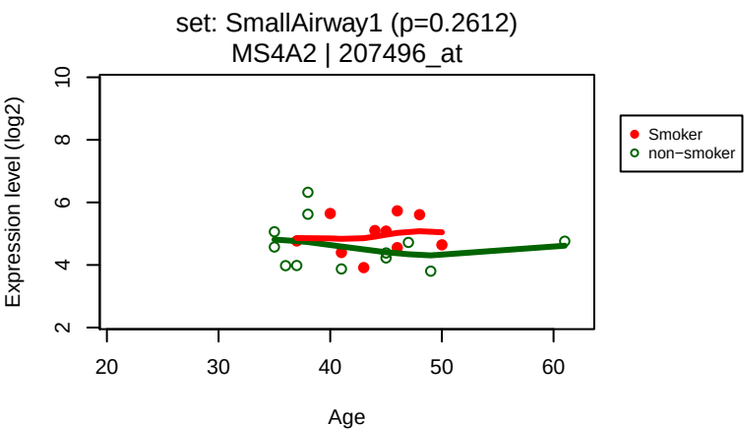
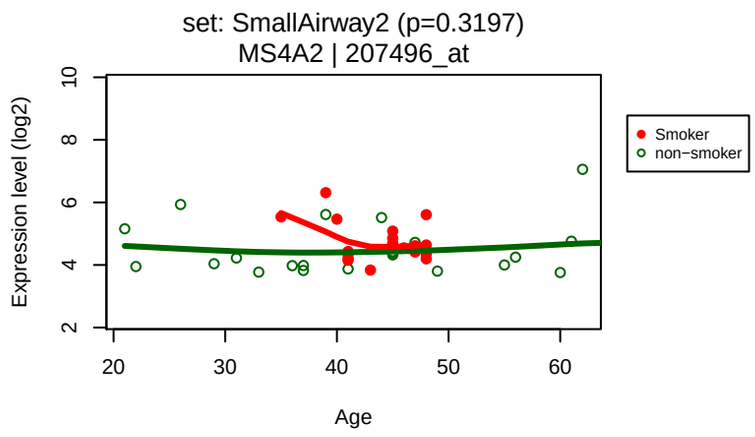
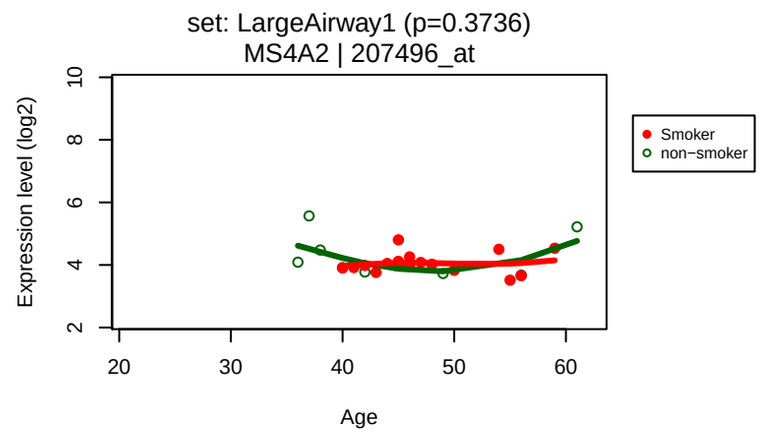

Gene expression of smokers and nonsmokers
gene: MS4A2 | 207496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2612
p=0.3197
p=0.5339
p=0.3736
p=0.8504
p=0.0172
p=0.5
p=0.0548
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

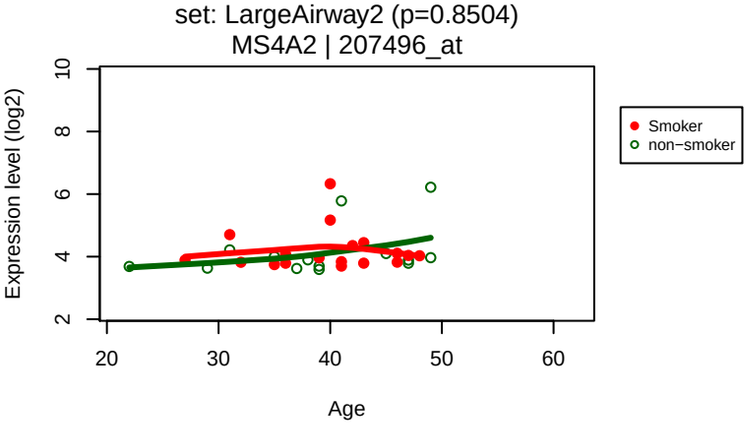

Gene expression ratio (smokers/nonsmokers)
gene: MS4A2 | 207496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2612
p=0.3197
p=0.5339
p=0.3736
p=0.8504
p=0.0172
p=0.5
p=0.0548
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

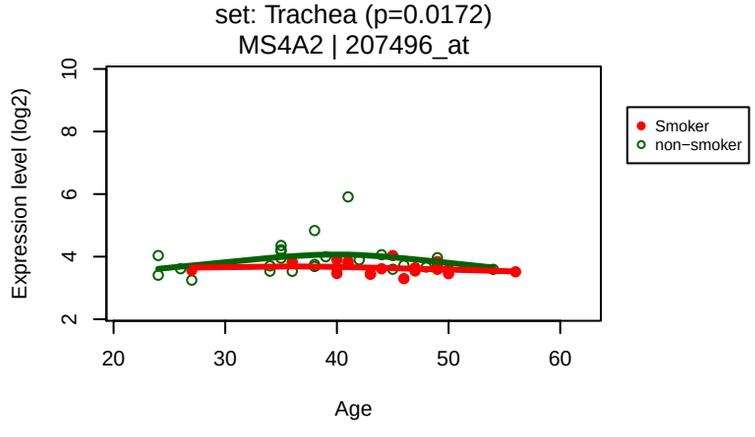

Gene expression ratio in Males
gene: MS4A2 | 207496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0672
p=0.8661
p=0.0027
p=0.3895
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A2 | 207496_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4523
p=0.612
p=0.271
p=0.0554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

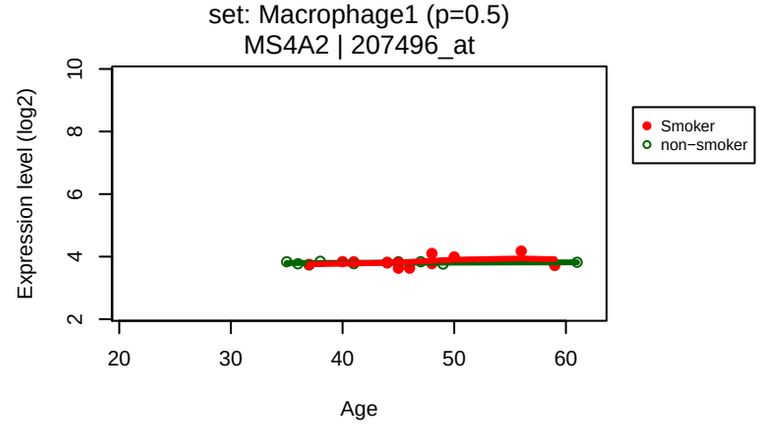

set: SmallAirway1 (p=0.2612)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker

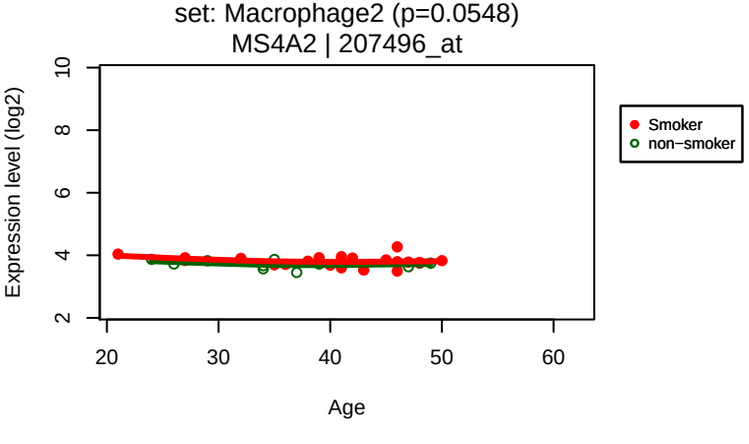

set: SmallAirway2 (p=0.3197)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker

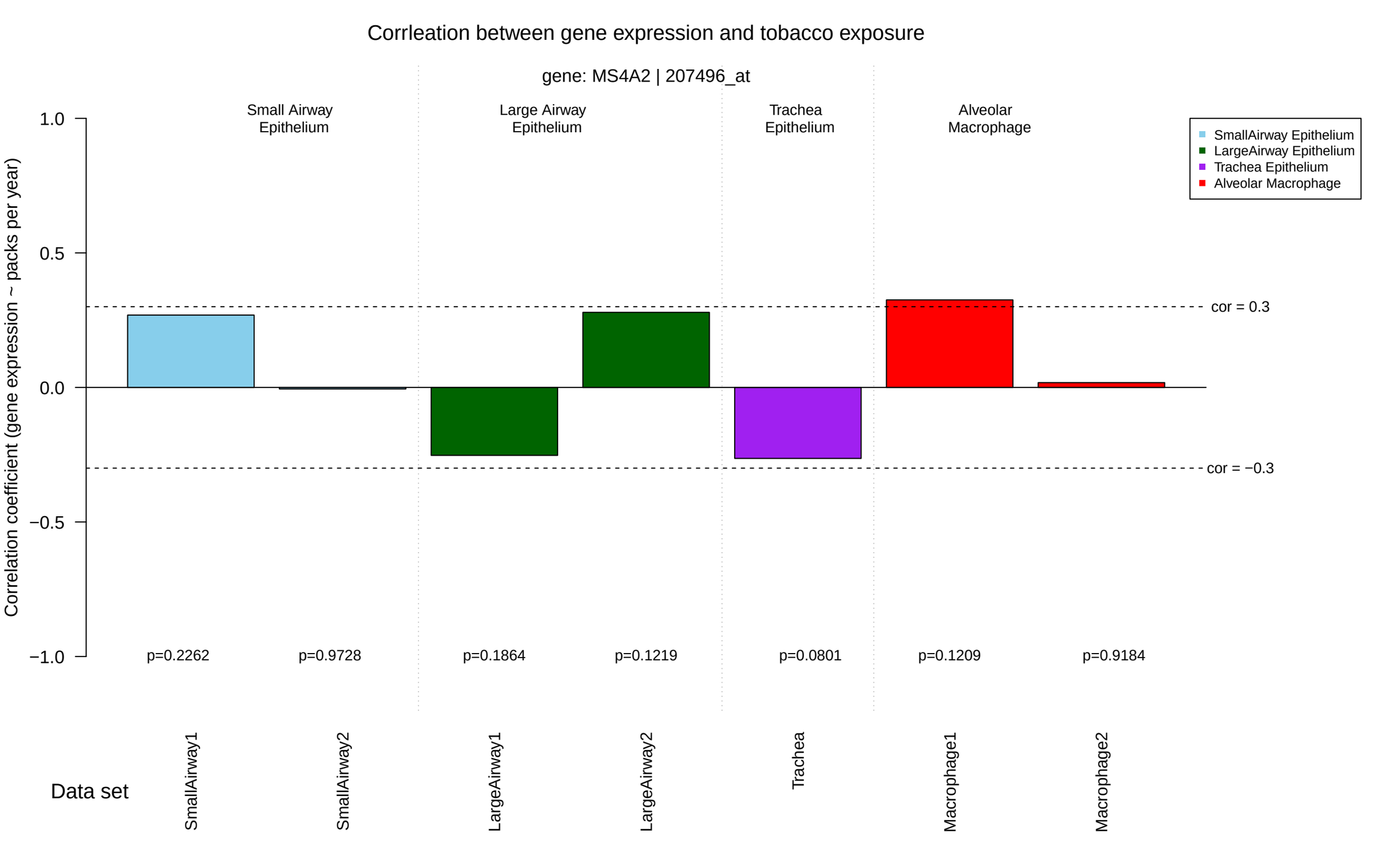

set: LargeAirway1 (p=0.3736)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8504)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0172)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0548)
MS4A2 | 207496_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A2 | 207496_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2262
p=0.9728
p=0.1864
p=0.1219
p=0.0801
p=0.1209
p=0.9184
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2